```
# Set up the packages and read the data
library(tidyverse)

fmr <- read_csv(
  "fmr2027.csv",
  col_types = cols(
    fips = col_character(),
    stusps = col_character(),
    state = col_character(),
    countyname = col_character(),
    county_town_name = col_character(),
    hud_area_code = col_character(),
    hud_area_name = col_character(),
    metro = col_integer(),
    pop2024 = col_double(),
    bedrooms = col_integer(),
    fmr = col_double()
  )
)

# Prepare the variables used in the graphs
fmr <- fmr |>
  mutate(
    metro = factor(metro, levels = c(0, 1),
                   labels = c("nonmetro", "metro")),
    record_type = if_else(str_ends(fips, "99999"),
                          "county code", "sub-county code")
  )

# Each HUD area has one rent for each bedroom size. Keep one of each.
area_rents <- fmr |>
  distinct(hud_area_code, hud_area_name, metro, bedrooms, fmr)

area_2br <- area_rents |>
  filter(bedrooms == 2)

records_2br <- fmr |>
  filter(bedrooms == 2)
```

```
# A few checks from the first pass through the data

dim(fmr)
str(fmr)
colSums(is.na(fmr))
sum(duplicated(fmr))
fmr |> count(fips, bedrooms) |> filter(n > 1)
fmr |> count(fips) |> count(n)
fmr |> filter(bedrooms == 2, pop2024 == 0) |> count(stusps)
fmr |>
  group_by(hud_area_code, bedrooms) |>
  summarise(n_rents = n_distinct(fmr), .groups = "drop") |>
  filter(n_rents > 1)
```

```
# Distribution of two-bedroom Fair Market Rents

ggplot(area_2br, aes(x = fmr)) +
  geom_histogram(binwidth = 50, boundary = 0) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "count", caption = "Figure 1") +
  theme_minimal(12)
```

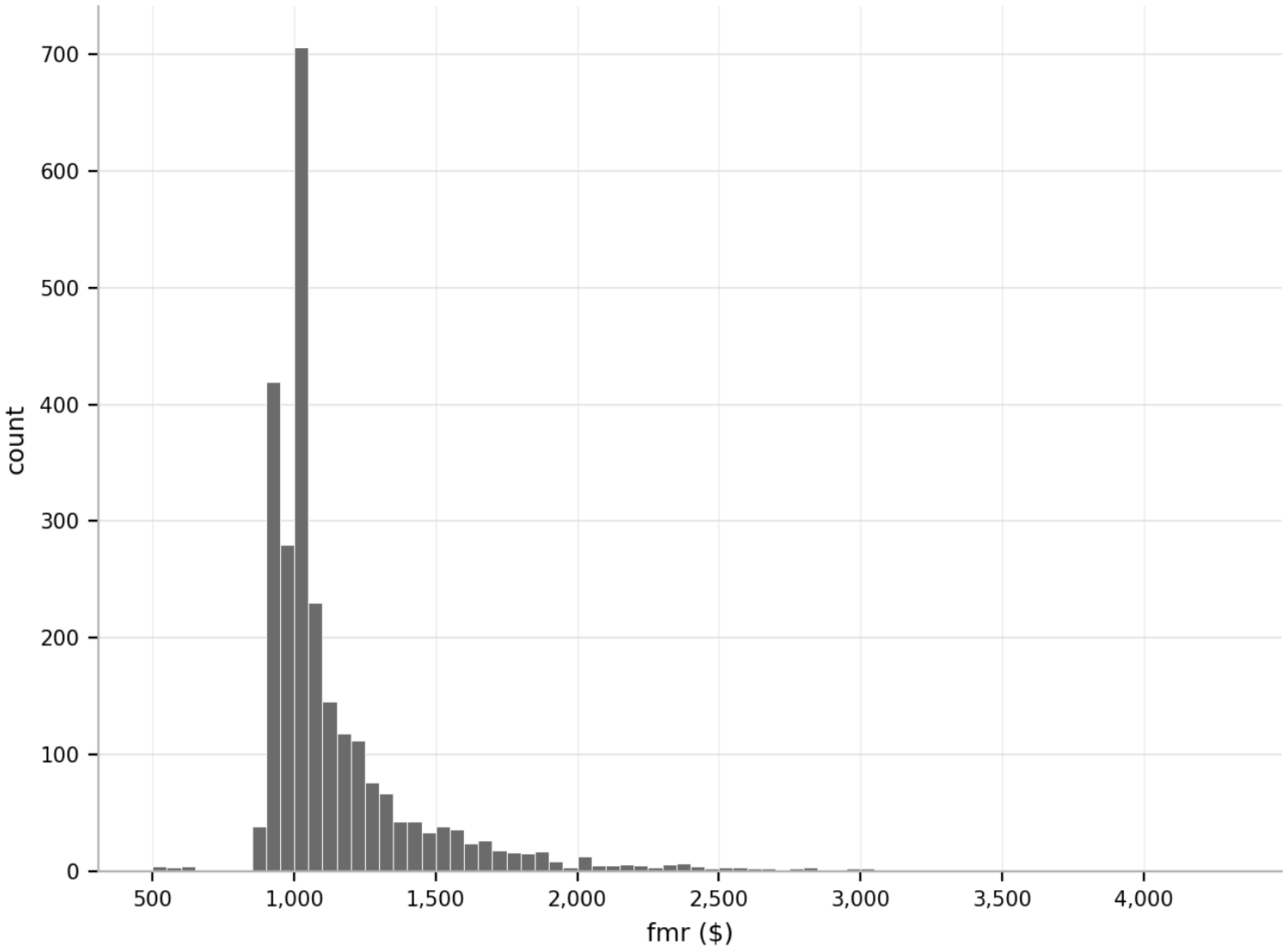

700
600
500
400
300
200
100
0
count
500
1,000
1,500
2,000
2,500
3,000
3,500
4,000
fmr ($)

Figure 1

```
# Fair Market Rent by bedroom count

ggplot(area_rents, aes(x = factor(bedrooms), y = fmr)) +
  geom_boxplot(outlier.alpha = 0.35, outlier.size = 0.7) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(x = "bedrooms", y = "fmr ($)", caption = "Figure 2") +
  theme_minimal(12)
```

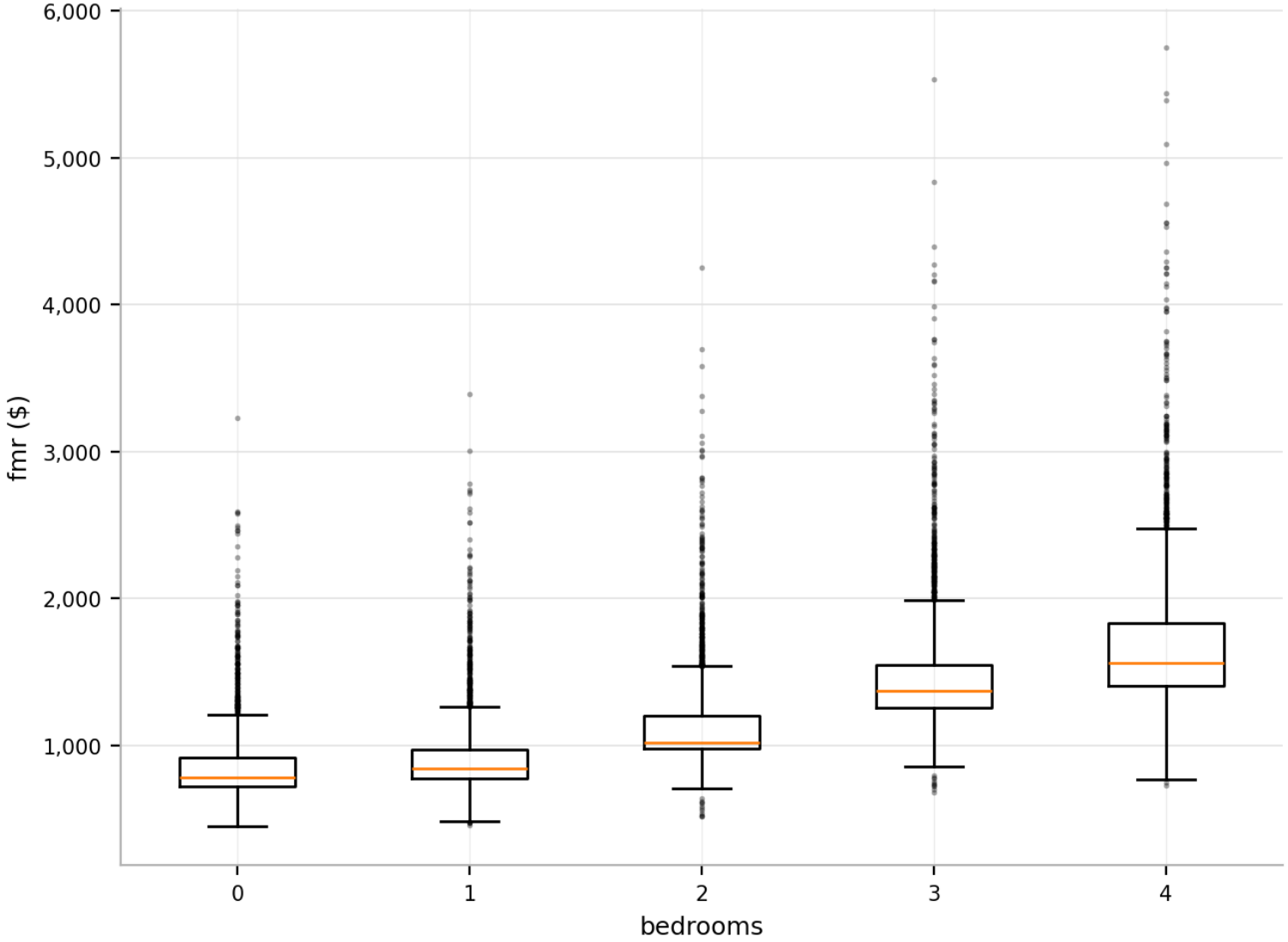


Figure 2

```
# Compare each bedroom size with the same area's two-bedroom rent

rent_wide <- area_rents |>
  select(hud_area_code, bedrooms, fmr) |>
  pivot_wider(names_from = bedrooms, values_from = fmr,
              names_prefix = "br")

bedroom_ratios <- rent_wide |>
  transmute(
    hud_area_code,
    `0` = br0 / br2,
    `1` = br1 / br2,
    `3` = br3 / br2,
    `4` = br4 / br2
  ) |>
  pivot_longer(
    cols = -hud_area_code,
    names_to = "bedrooms",
    values_to = "ratio"
  ) |>
  mutate(bedrooms = factor(bedrooms, levels = c("0", "1", "3", "4")))

ggplot(bedroom_ratios, aes(x = ratio)) +
  geom_histogram(binwidth = 0.01, boundary = 0) +
  facet_wrap(~bedrooms, ncol = 2) +
  labs(x = "fmr / fmr (2 bedrooms)", y = "count", caption = "Figure 3") +
  theme_minimal(12)
```

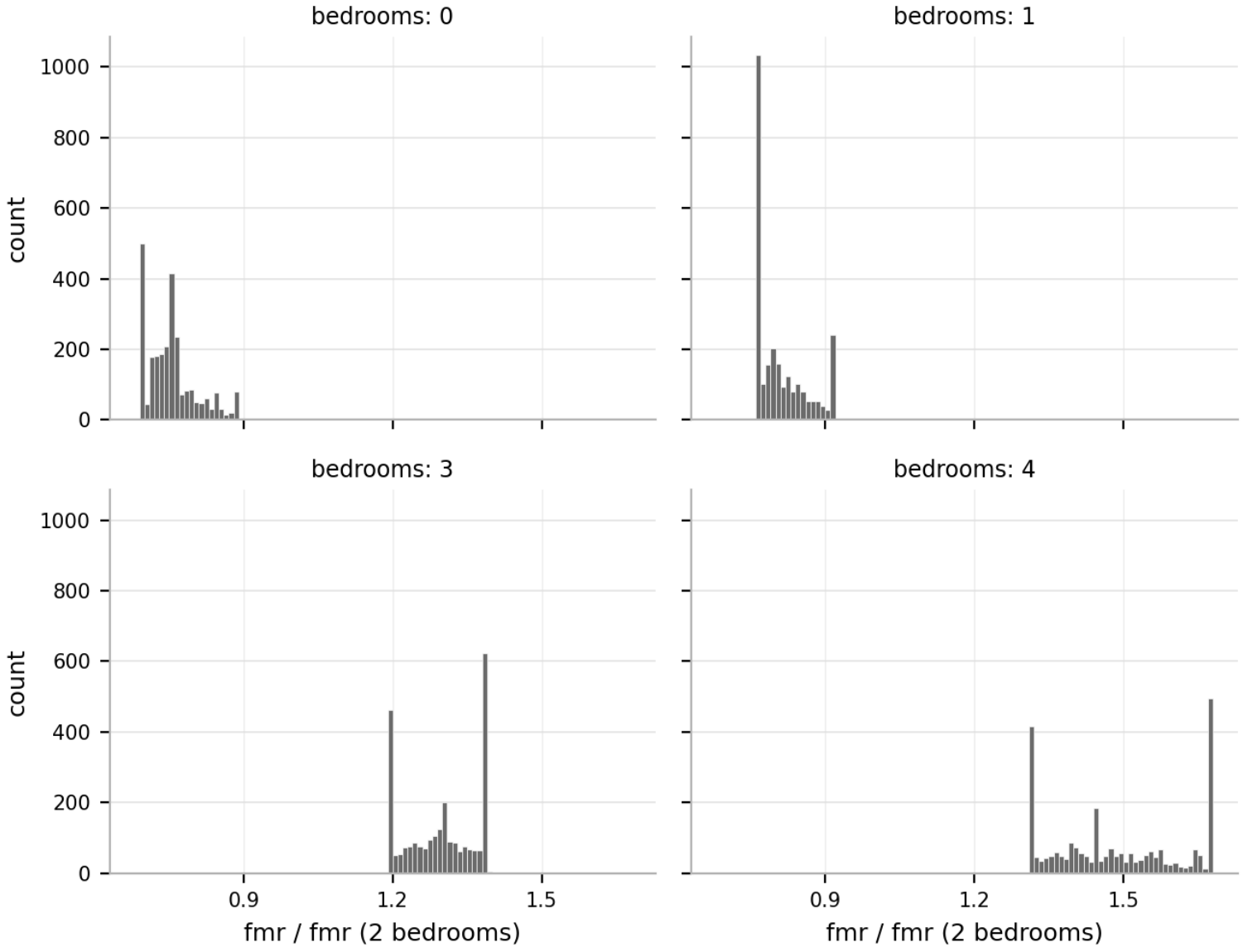

bedrooms: 0
bedrooms: 1
bedrooms: 3
bedrooms: 4
count
1000
800
600
400
200
0
0.9
1.2
1.5
fmr / fmr (2 bedrooms)

Figure 3

```
# Compare two-bedroom rents for metro and nonmetro HUD areas

ggplot(area_2br, aes(x = metro, y = fmr)) +
  geom_boxplot(outlier.alpha = 0.35, outlier.size = 0.7) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(x = "metro", y = "fmr ($)", caption = "Figure 4") +
  theme_minimal(12)
```

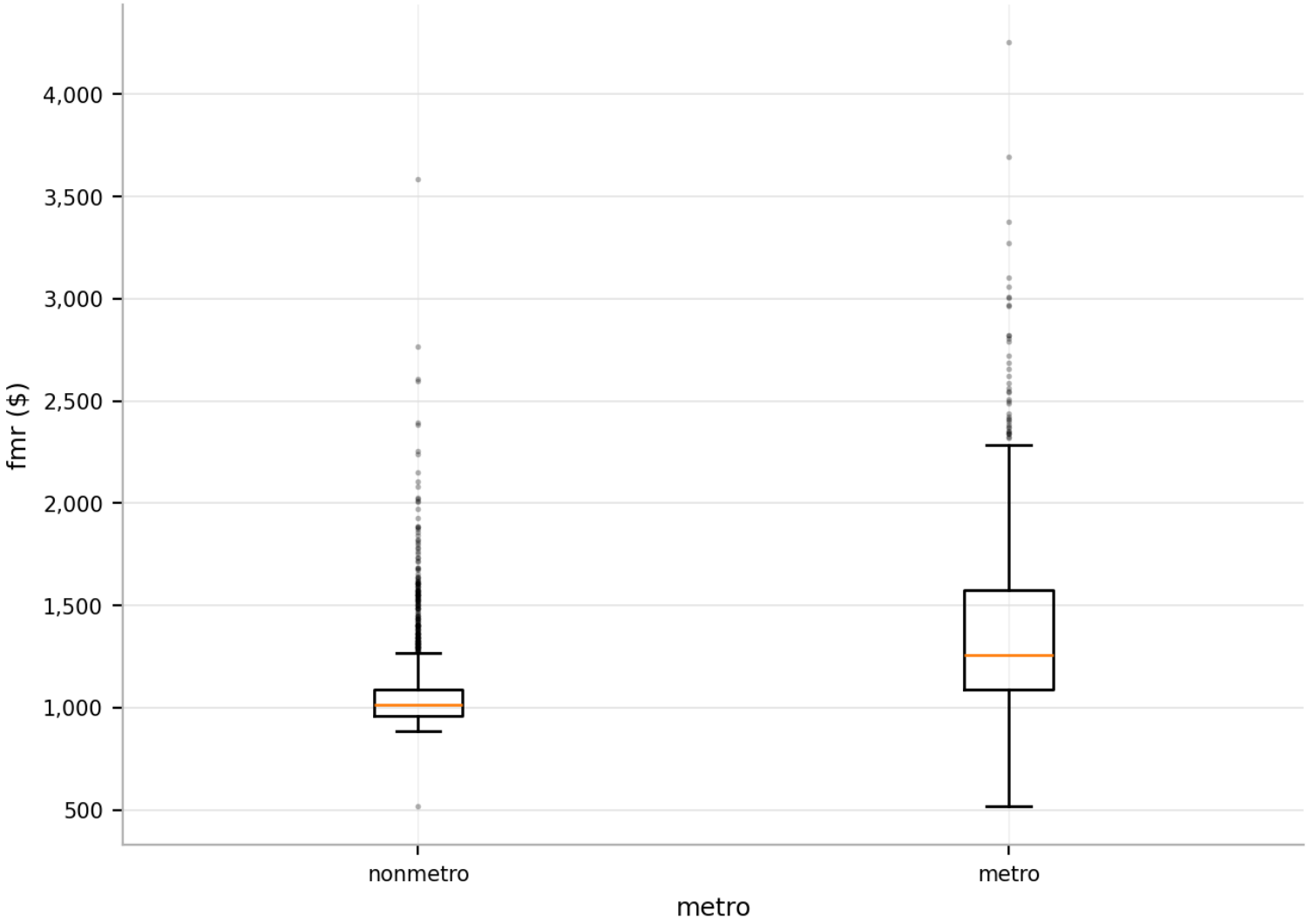

4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
fmr ($)
nonmetro
metro
metro

Figure 4

```
# Two-bedroom FMR across states; each point is a HUD area

state_areas <- records_2br |>
  distinct(hud_area_code, stusps, metro, fmr) |>
  mutate(stusps = fct_reorder(stusps, fmr, .fun = median))

ggplot(state_areas, aes(x = fmr, y = stusps)) +
  geom_point(
    aes(colour = metro),
    alpha = 0.65,
    size = 1.3,
    position = position_jitter(width = 0, height = 0.15, seed = 1)
  ) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "stusps", colour = "metro", caption = "Figure 5") +
  theme_minimal(12)
```

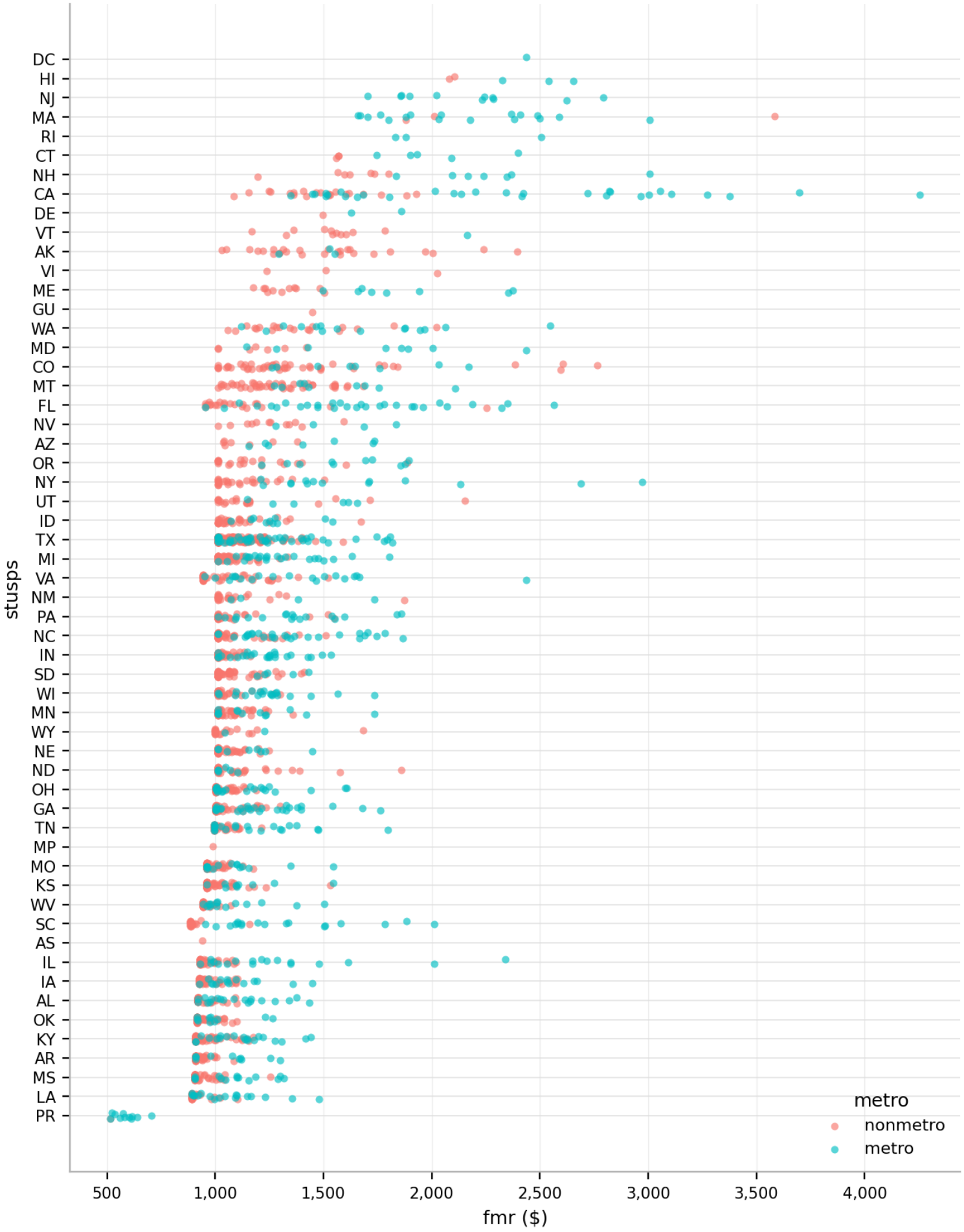

stusps
DC
HI
NJ
MA
RI
CT
NH
CA
DE
VT
AK
VI
ME
GU
WA
MD
CO
MT
FL
NV
AZ
OR
NY
UT
ID
TX
MI
VA
NM
PA
NC
IN
SD
WI
MN
WY
NE
ND
OH
GA
TN
MP
MO
KS
WV
SC
AS
IL
IA
AL
OK
KY
AR
MS
LA
PR
500
1,000
1,500
2,000
2,500
3,000
3,500
4,000
fmr ($)
metro
nonmetro
metro

Figure 5

```
# Population and two-bedroom FMR, separated by record type
# Zero-population records are shown in Figure 8; a log axis cannot show zero.

ggplot(filter(records_2br, pop2024 > 0), aes(x = pop2024, y = fmr)) +
  geom_point(aes(colour = metro), alpha = 0.45, size = 1.2) +
  facet_wrap(~record_type) +
  scale_x_log10(breaks = c(10, 1000, 100000, 10000000),
                labels = scales::label_comma()) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(x = "pop2024", y = "fmr ($)", colour = "metro", caption = "Figure 6") +
  theme_minimal(12)
```

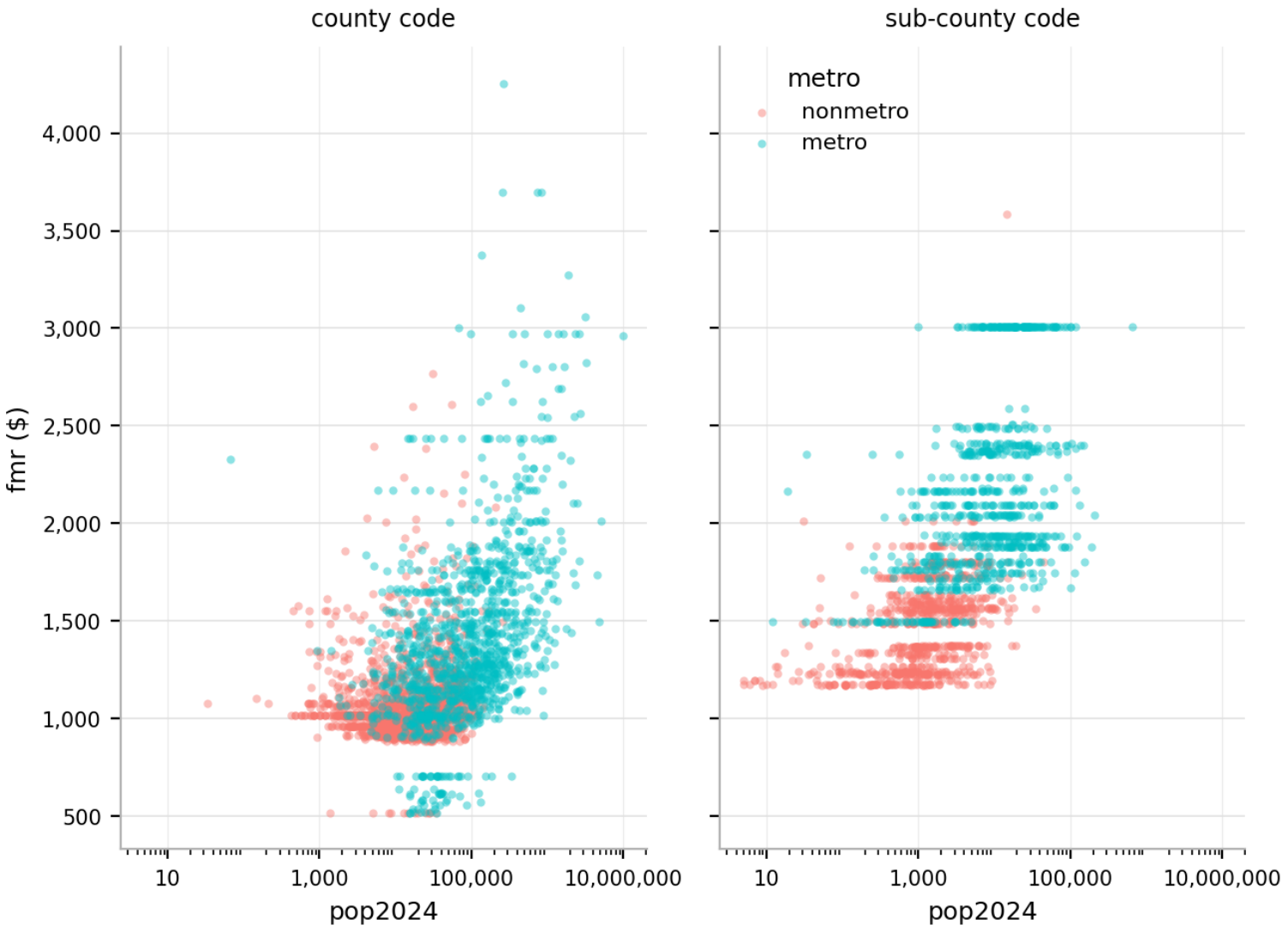

county code
sub-county code
metro
nonmetro
metro
fmr ($)
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
10
1,000
100,000
10,000,000
pop2024

Figure 6

```
# Missing town names by record type

name_status <- records_2br |>
  mutate(county_town_name = if_else(
    is.na(county_town_name), "missing", "present"
  ))

ggplot(name_status, aes(x = record_type, fill = county_town_name)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::label_percent()) +
  labs(x = "record_type", y = "proportion", fill = "county_town_name",
       caption = "Figure 7") +
  theme_minimal(12)
```

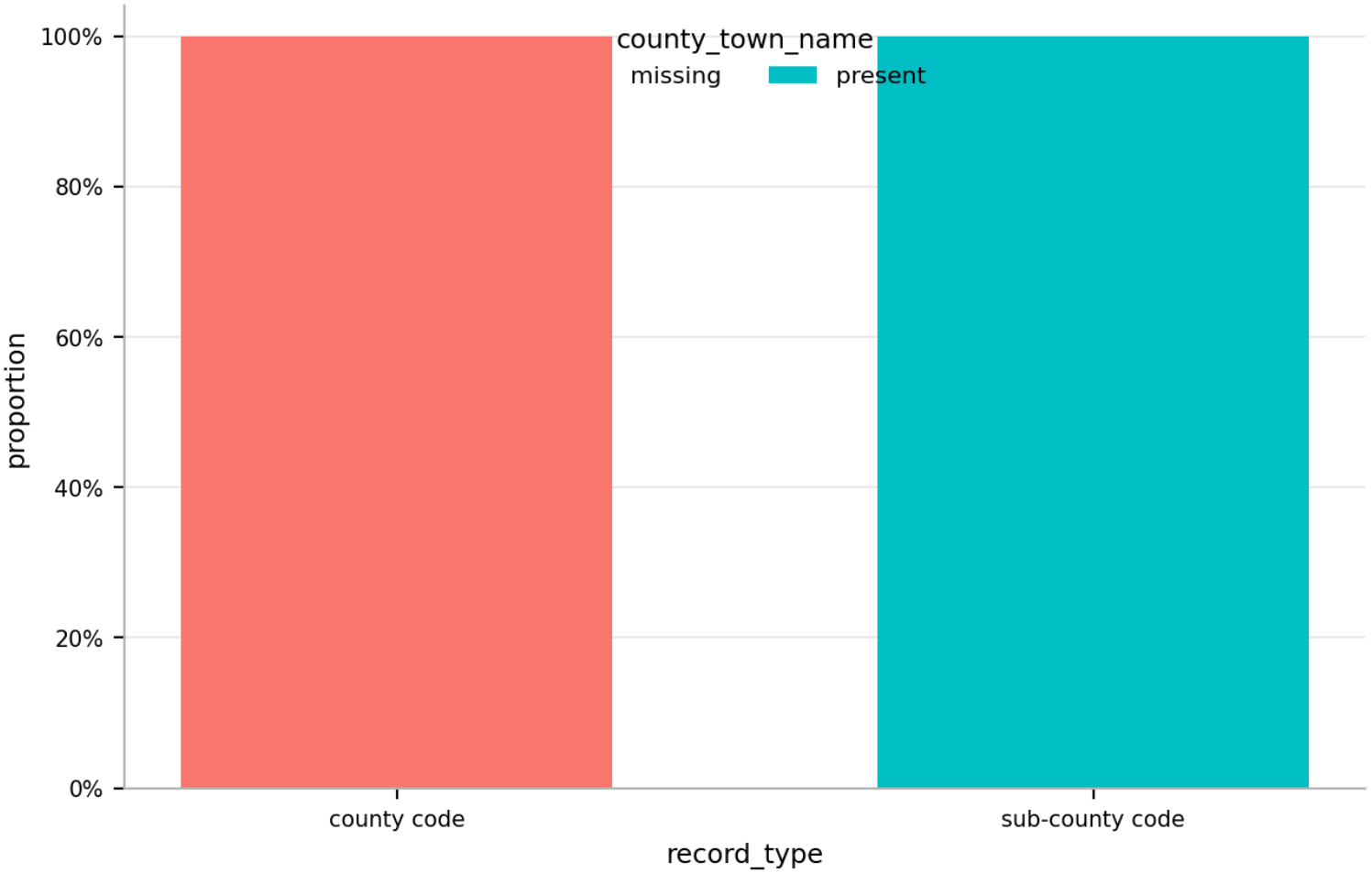

county_town_name
missing
present
100%
80%
60%
40%
20%
0%
proportion
county code
sub-county code
record_type

Figure 7

```
# Geographic records with zero population

zero_pop <- records_2br |>
  filter(pop2024 == 0) |>
  count(stusps) |>
  mutate(stusps = fct_reorder(stusps, n))

ggplot(zero_pop, aes(x = n, y = stusps)) +
  geom_col() +
  labs(x = "count", y = "stusps", caption = "Figure 8") +
  theme_minimal(12)
```

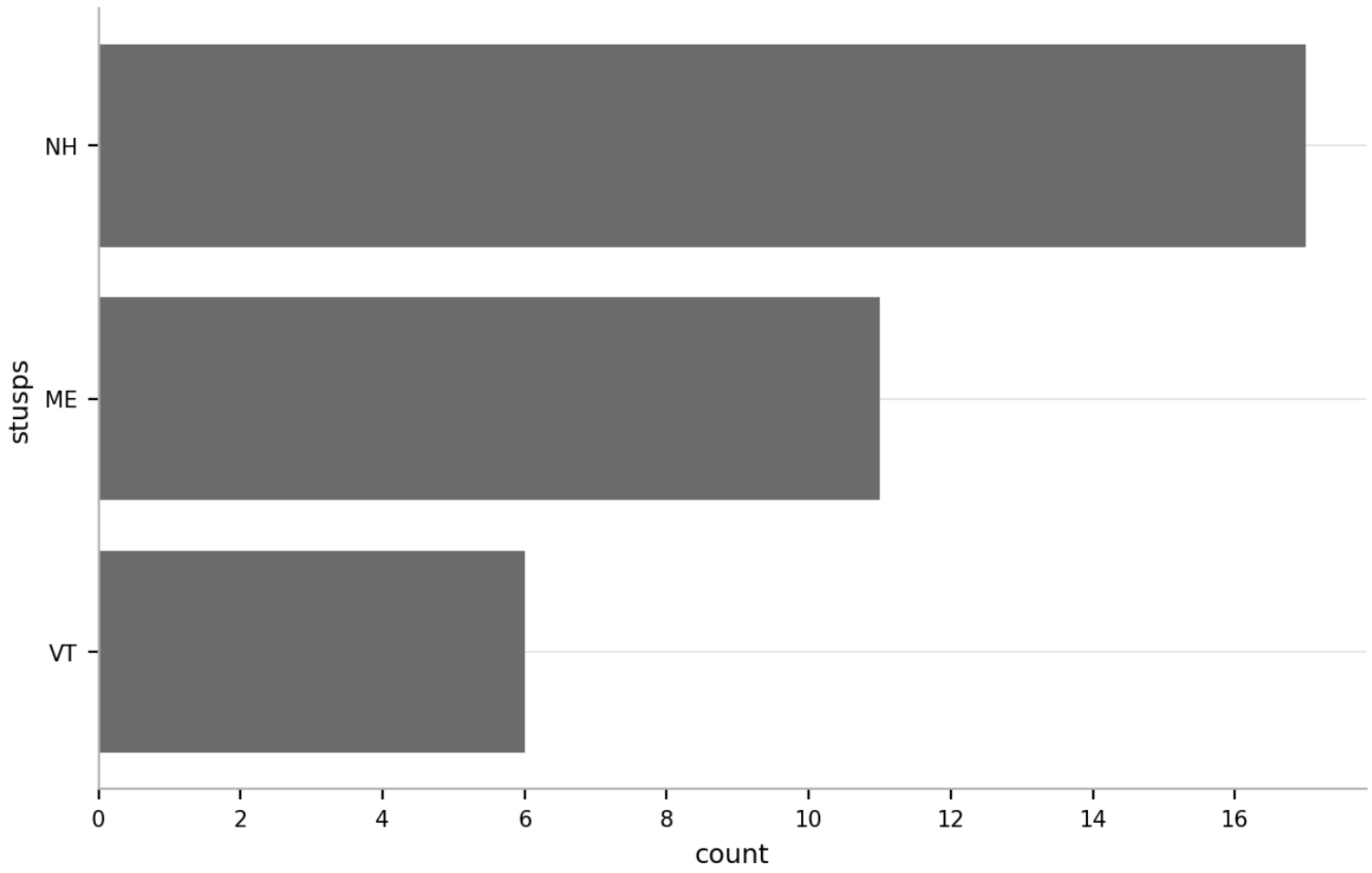


Figure 8

```
# Geographic records and distinct HUD areas by state

state_counts <- records_2br |>
  group_by(stusps) |>
  summarise(
    records = n_distinct(fips),
    hud_areas = n_distinct(hud_area_code),
    .groups = "drop"
  ) |>
  pivot_longer(
    cols = c(records, hud_areas),
    names_to = "unit",
    values_to = "count"
  ) |>
  mutate(stusps = fct_reorder(stusps, count, .fun = max))

ggplot(state_counts, aes(x = count, y = stusps)) +
  geom_line(aes(group = stusps), colour = "grey70") +
  geom_point(aes(colour = unit), size = 1.8) +
  scale_x_log10(breaks = c(1, 3, 10, 30, 100, 300)) +
  labs(x = "count", y = "stusps", colour = "unit", caption = "Figure 9") +
  theme_minimal(12)
```

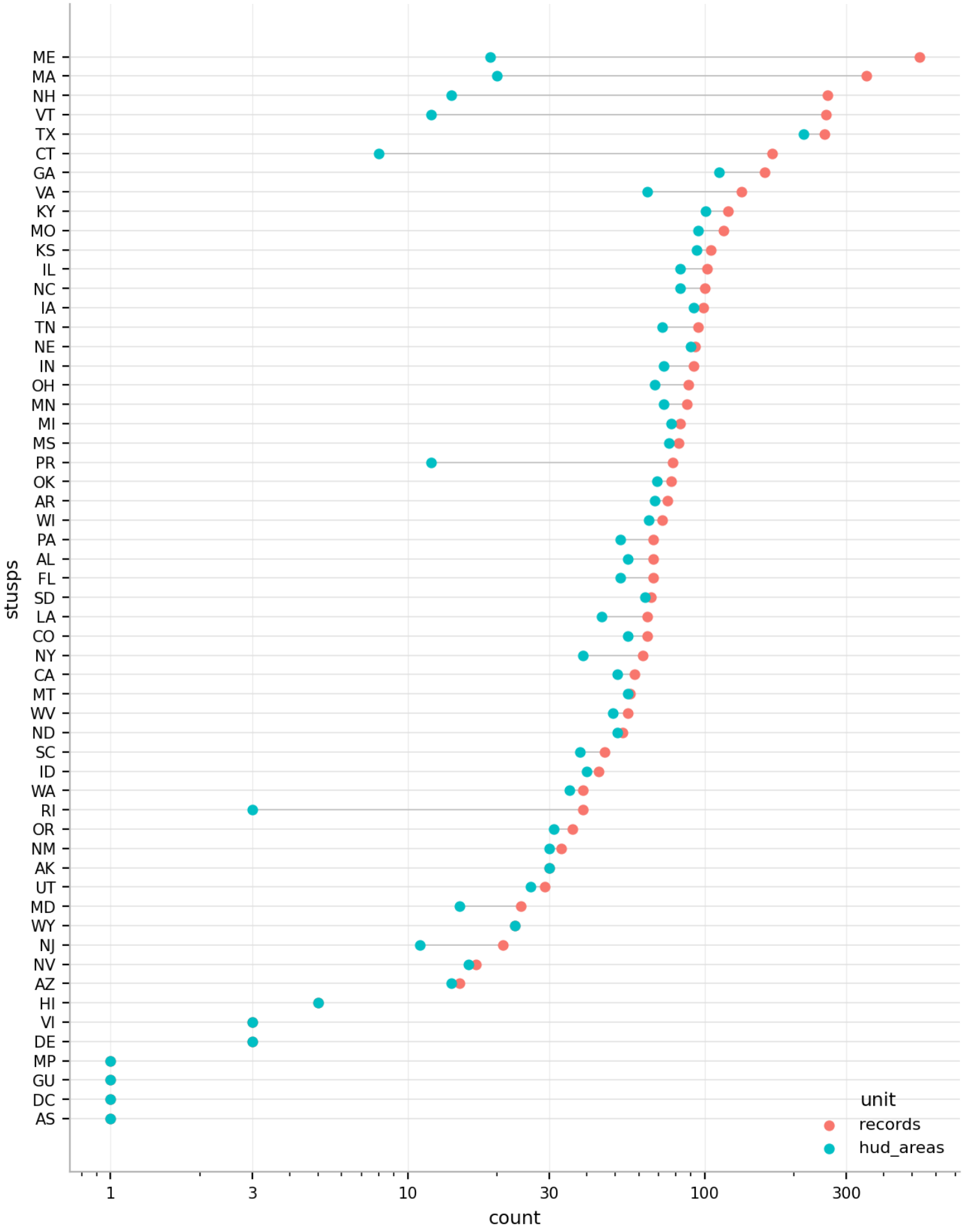

ME
MA
NH
VT
TX
CT
GA
VA
KY
MO
KS
IL
NC
IA
TN
NE
IN
OH
MN
MI
MS
PR
OK
AR
WI
PA
AL
FL
SD
LA
CO
NY
CA
MT
WV
ND
SC
ID
WA
RI
OR
NM
AK
UT
MD
WY
NJ
NV
AZ
HI
VI
DE
MP
GU
DC
AS
stusps
1
3
10
30
100
300
count
unit
records
hud_areas

Figure 9

```
# Highest and lowest two-bedroom rents, with each area's full rent schedule

extreme_codes <- bind_rows(
  area_2br |> arrange(fmr) |> slice_head(n = 10),
  area_2br |> arrange(fmr) |> slice_tail(n = 10)
) |>
  pull(hud_area_code)

extreme_names <- area_2br |>
  filter(hud_area_code %in% extreme_codes) |>
  arrange(fmr) |>
  pull(hud_area_name) |>
  str_remove(" HUD Metro FMR Area")

extreme_areas <- area_rents |>
  filter(hud_area_code %in% extreme_codes) |>
  mutate(
    hud_area_name = str_remove(hud_area_name, " HUD Metro FMR Area"),
    hud_area_name = factor(hud_area_name, levels = extreme_names),
    bedrooms = factor(bedrooms)
  )

ggplot(extreme_areas, aes(x = fmr, y = hud_area_name, colour = bedrooms)) +
  geom_point(size = 1.8) +
  scale_x_continuous(labels = scales::label_comma()) +
  labs(x = "fmr ($)", y = "hud_area_name", colour = "bedrooms",
       caption = "Figure 10") +
  theme_minimal(12)
```


Santa Cruz-Watsonville, CA MSA
San Francisco, CA
Nantucket County, MA
Napa, CA MSA
San Jose-Sunnyvale-Santa Clara, CA
Santa Maria-Santa Barbara, CA MSA
Santa Ana-Anaheim-Irvine, CA
Boston-Cambridge-Quincy, MA-NH
San Benito County, CA
New York, NY
Caguas, PR
Aguadilla, PR MSA
Mayagüez, PR
San Germán, PR
Ponce, PR
Arecibo, PR MSA
Yauco, PR
Barranquitas-Aibonito, PR
Guayama, PR MSA
Puerto Rico HUD Nonmetro Area
hud_area_name
1,000
2,000
3,000
4,000
5,000
6,000
fmr ($)
bedrooms
0
1
2
3
4

Figure 10